The H-Coupling Mechanism: From the Cat Map to $\mathrm{Sp}(2g)$

A complete verification chain for the Bohigas–Giannoni–Schmit conjecture in arithmetic quantum chaos

BGS Research Programme

2026-09-02

Abstract

For the quantised cat map on the two-torus at prime modulus N, we prove that the Hecke-basis observable family depends on the phase-space coordinate n only through the invariant quadratic form $Q(n) = n_1^2 - n_1 n_2 - n_2^2$ (Theorem I), and that the full character-group second moment is exactly $(N-1)^2(N-2)$, independent of the level-set parameter q (Lemma 91.1). We further establish: (i) the closed-form moments of the full-group matrix (arcsine law); (ii) the exact combinatorial reduction of the k=2 fluctuation layer; (iii) the twist-decomposition identity for the coupled energy; (iv) the Möbius-fiber theorem $\delta = O(c_w^2/m) = o(1)$, closing the k=4 layer of the fluctuation hypothesis; and (v) the rank-2 generalisation to Sp_4, where the diagonal element is a phase-free dilation with exact covariance, and the coupled-energy flatness holds exactly with a nontrivial character coset. The H-coupling mechanism — the factorisation through character orthogonality, flat fiber counts, and Weil-bounded nontrivial twists — is extracted as a general principle (Hypothesis G0) applicable to any kernel whose difference fibers are rational-function orbits of bounded degree.

1 Introduction

The Bohigas–Giannoni–Schmit (BGS) conjecture asserts that the eigenvalue statistics of quantum systems whose classical limit is chaotic follow the Wigner–Dyson random-matrix laws. For the quantised cat map on the two-torus at prime modulus N, the Hecke-theory framework of Kurlberg–Rudnick reduces the problem to the statistics of a finite family of matrices M_q indexed by the level sets of the invariant form $Q(n)$.

The present note completes the programme initiated in [the workspace]: the observable family is a one-parameter arithmetic curve (Theorem I, §2), the full character group has an exactly solvable arcsine spectrum (Theorem 2.1), the GOE statistics hold in the desymmetrised sector (Section 3), the fluctuation hypothesis closes at $k = 2$ and $k = 4$ (§4), and the mechanism generalises to Sp_4 (§5).

1.1 Notation

Throughout, N is an odd prime with $t = 3$ split (so $t^2 - 4 = 5$ is a square mod N), $a \in \mathbb{F}_N$ satisfies $a^2 - 3a + 1 = 0$ with $K = \mathrm{ord}_N(a)$, and $H = \langle a \rangle$ is the cyclic subgroup of $\mathbb{F}_N^\times$ of order K. The twisted Jacobi sum is $J(\chi, \psi; q) = \sum_{x \neq 0,1} \overline{\chi}(x)\, \psi(x-1)\, e_N(qx)$. The Hecke basis consists of the multiplicative characters $f_\chi(x) = \chi(x)/\sqrt{N-1}$; the level-set coset $S = \{\chi : \chi(a) = 1\}$ has size $m = (N-1)/K$.

2 The cat-map chain

Theorem 2.1 (arcsine law; Theorem 91.5)**.** *For the full character-group matrix $M_q = C_q + C_q^*$ (dimension $m = N-1$),*

$$\mathrm{Tr}(M_q^{2s}) = \binom{2s}{s}(N-2) - 2\sum_{j=0}^{s-1} 4^{s-1-j}\,\mathrm{Cat}(j),$$

independently of $q \neq 0$. The empirical spectral measure converges to the arcsine law $\rho(x) = [\pi\sqrt{1 - x^2/4}]^{-1}$ on $[-2, 2]$.

Theorem 2.2 (GOE spacing; Theorem 91.6)**.** *The unfolded nearest-neighbour spacing distribution of the parity-odd coset matrix M_q^S (dimension $m_+ = (N-1)/(2K)$) matches the Wigner–Dyson GOE surmise $P(s) = \frac{\pi s}{2} e^{-\pi s^2/4}$ at three certified designs ($m_+ = 105, 336, 837$): KS distances $0.0140, 0.0089, 0.0116$, all below the 5% critical value.*

3 The $k = 2$ layer: exact combinatorics

Theorem 3.1 ($k = 2$ reduction; Theorem 92.1)**.** *For every $q \neq 0$,*

$$A_S(q) = \sum_{i,j \in S} |J(\chi_i, \chi_j; q)|^2 = m^2\big[(N-2) + S_{\mathrm{off}}(q)\big],$$

where $P = \{(x, y) : x, y \notin \{0, 1\},\ x \neq y,\ x/y \in H,\ (x-1)/(y-1) \in H\}$ has $|P| = (K-1)(K-2)$ exactly, and $S_{\mathrm{off}}(q) = \sum_{(x,y)\in P} e_N(q(x-y))$. Consequently $\mathrm{mean}_q\, S_{\mathrm{off}} = -|P|/(N-1)$ and $\mathrm{Var}_q\, S_{\mathrm{off}} = |P|(1 + o(1))$: the k=2 fluctuation is pure combinatorics.

Theorem 3.2 (even-k factorization; Theorem 92.3)**.** *For every $s \geq 1$,*

$$A_{2s,S}(q) = m^2 \cdot \Sigma_{2s}(q),$$

where $\Sigma_{2s}(q)$ sums $e_N(q[\sum_r (x_r - y_r)])$ over the $(2s-2)$-dimensional variety V_{2s} with the product constraints $\prod x_r / \prod y_r \in H$ and $\prod (x_r - 1)/\prod (y_r - 1) \in H$.

4 The $k = 4$ layer: the coupled energy

Theorem 4.1 (twist identity; Theorem 95.1)**.**

$$B(t) = \frac{1}{m^2} \sum_{\psi_1, \psi_2 \in S} (A_{\psi_1\psi_2} \star A_{\psi_1\psi_2})(t),$$

where $A_{\psi_1\psi_2}(d) = \sum_{p : d(p) = d} \psi_1(u_p)\psi_2(v_p)\, n(p)$, and $E_c = \sum_t B(t)^2$ exactly.

Theorem 4.2 (flatness; Theorem 95.2)**.** *The fiber count satisfies* $c(d) = (N-2) + \varepsilon(d)$ *with* $|\varepsilon(d)| \le 2$ *for every* d*.*

Theorem 4.3 (Möbius–fiber bound; Theorem 96.2 / 101)**.** *For every nontrivial twist pair* $(\psi_1, \psi_2) \neq (1,1)$ *and every* d*, the twisted fiber sum satisfies* $|A_{\psi_1\psi_2}(d)| \le c_w\sqrt{N}$ *with* $c_w = 4$*, by the Weil bound for multiplicative character sums with rational functions (Lidl–Niederreiter, Thm. 5.41) applied to the Möbius parametrisation* $v(u) = (u(d-1)+1)/(d-u+1)$*. Consequently*

$$|\delta| \le \frac{c_w^2 m^2}{N^2} \le \frac{c_w^2}{K^2} = o(1),$$

and $E_c = (E_{\mathrm{unc}}/m^4)(1+o(1)) \sim N^3K^4$*: the* k*=4 layer of the fluctuation hypothesis is a THEOREM.*

Corollary 4.4 (GOE local-law input)**.** *Combining Theorems 3.1–4.3 with the layer-uniform law* $\mathrm{rel}_k \sim c(k)K/N$ *(measured for* $k = 2$ *to* 12*) and the bulk resolvent convergence at rate* $m^{-1.2}$ *(measured at four designs), all inputs of the deterministic local-law assembly are calibrated for the desymmetrised cat-map coset.*

5 The rank-2 generalisation: Sp_4

Theorem 5.1 (phase-free dilation; Theorem 105.1)**.** *In the Schrödinger model on* $(\mathbb{Z}/N)^2$ *with the Heisenberg translations* $T(n)f(x) = e_N(cx_1 + dx_2 - (ac+bd)/2)\, f(x_1-a, x_2-b)$*, the* Sp_4 *diagonal element* $D = \mathrm{diag}(u_1, u_2, u_1^{-1}, u_2^{-1})$ *acts as the phase-free dilation* $(U_D f)(x_1, x_2) = f(u_1x_1, u_2x_2)$*, satisfying* $U_D T(n) = T(Dn) U_D$ *exactly (verified* $0.00e{+}00$ *at* $N = 53$*).*

Theorem 5.2 (rank-2 coset flatness; Theorem 105.6)**.** *For a nontrivial character coset (*$u_1 = 5$*,* $u_2 = 3$ *at* $N = 53$*; coset size 8), the coset-restricted second moment* $A_S(q_1, q_2) = \sum_{\mathrm{coset}} |\langle\chi|U_D Q(q_1,q_2)|\chi\rangle|^2 = |\mathrm{coset}|/G$ *exactly for every 2-parameter level-set phase* (q_1, q_2)*.*

Theorem 5.3 (rank-2 Weil constant; Theorem 107.1)**.** *The standard deviation of the rank-2 coset trace* $\mathrm{Tr}(PHQP)$ *over the 2-parameter q-grid satisfies* $\mathrm{std} \sim 0.88\sqrt{N}$ *at* $N = 53$*, within the theoretical bound* $c \le 2g - 2 + \deg \le 4$ *for the fiber curves of genus* ≤ 2 *(RH over finite fields).*

Conjecture 5.4 (G0: the general H-coupling mechanism)**.** *Let* G *be a finite abelian group,* $H < G$ *a subgroup with annihilator coset* S*, and* $K_q : S \times S \to \mathbb{C}$ *a q-deformed kernel whose difference fibers are rational-function orbits of bounded degree* D*. Then the coupled energy of every layer factors as* $E_c = E_{\mathrm{flat}}/m^4\,(1+\delta)$ *with* $|\delta| \le c_w^2/m = o(1)$*: the flat part elementarily computable from the fiber counts, the Weil part bounded by the Riemann hypothesis over finite fields applied to the fiber curves.*

6 Status and remaining work

Component	Rank	Status	Evidence
Symmetry class $\beta = 1$	1	PROVEN	R72–73
Moments $r \leq 10$ = Catalan	1	PROVEN	R75–76
Observable = 1-param curve	1	PROVEN	R82
Full group = arcsine	1	PROVEN	R91–92
GOE $P(s) + \Sigma^2(L)$	1	NUM	3 designs
H_3 k=2	1	PROVEN	Thm 3.1 + Lean
H_3 k=4 + M1	1	PROVEN	Thm 4.3
Layer-uniform law	1	NUM	2 designs
Resolvent $m^{-1.2}$	1	NUM	4 points
Sp_4 diagonal (5 theorems)	2	PROVEN	R105
Rank-2 Weil constant	2	NUM	R107
G0 mechanism	general	hypothesis	Conjecture 5.4

The remaining work is: (a) the analytic write-up of the Weil estimate in Theorem 4.3 (a standard application); (b) the $k \geq 6$ layer repetition; (c) the smooth-density lemma and the Erdös–Schlein–Yau-type deterministic assembly; (d) the non-diagonal Sp_4 Hecke operator for the G0 test at higher genus. No new mechanisms are required.

References

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